

DSA 210 Project Proposal

Title: The Impact of Poverty and Ethnic Composition on Mortality Rates Across U.S. Counties

Research Question:

How do poverty levels and ethnic composition affect mortality rates across U.S. counties?

This project aims to analyze how socioeconomic level and ethnicity influence mortality rates at the county level in the United States. In particular, the study focuses on the relationship between poverty levels, ethnic composition, and mortality outcomes with the goal of identifying patterns and disparities across regions.

The primary data source for poverty level and ethnicity is the U.S. Census Bureau, which is the official statistical agency of the United States and provides highly reliable, standardized, and comprehensive county oriented data. This source includes detailed information on poverty rates and population composition by ethnicity. Mortality data is obtained from Kaggle, where datasets compiled from official health statistics are publicly available. All datasets will be downloaded in CSV and Excel formats.

The dataset will include observations for approximately 3,000 U.S. counties over a five-year period, depending on data availability and overlap across sources. The variables that being considered is poverty rate ,ethnicity (such as white, Asian, Hispanic americans) and mortality rate (deaths per 100,000 individuals). The data is numerical with categorical identifiers such as county and state codes.

Data collection will involve downloading the datasets and merging them using common counties and years . In the implementation part, the organization of data which includes cleaning or removing irrelevant information, the part of data chosen according to overlapping the same year and county is being considered.

Finally, this project is expected to provide insights into how economic disadvantage and demographic structure are associated with mortality rates across U.S. counties.